

POAMS: LECTURE 6
UNIFORM MOTION: SOME CONSEQUENCES
OF TIME-DILATION AND THE 'CONE MODEL'

Having derived the time-dilation formula for uniform motion and the associated 'cone model' in our Normal Realist alternative to Einstein's Special Relativity, in this lecture I will examine some of the consequences of these, including a solution to the 'EPR paradox', the Doppler effect, and a solution to the so-called 'twins paradox'. The lecture is based on some of 'Light Speed...' Chapter 3.

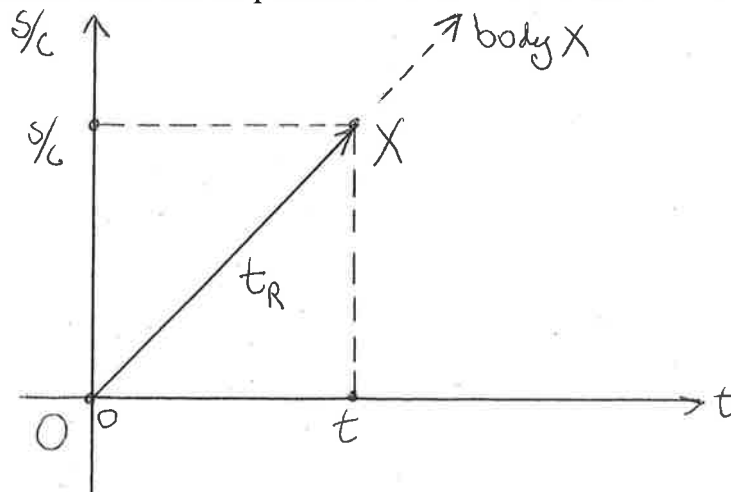
1. Time-dilation in uniform motion

As a reminder from the last lecture, our Normal Realist postulates for the study of *uniform motion* are:

Postulate 1: *The laws of Physics are the same in all inertial frames.*

Postulate 2: *Observational distance and time have a constant ratio of units, c , for all inertial observers sharing that same conventional choice of units.*

Consider the case of a body X in uniform motion relative to an observer O , as studied in the last lecture. Body X travels a distance s measured in metres, in a time t , recorded in seconds by X 's clock, relative to O . In keeping with Postulate 2, this distance travelled by X relative to O can be recorded in seconds by the use of the conversion factor c . We then have the *two-dimensional time* diagram, as shown below. In the figure, the two axes (labelled s/c and t) are both measured in units of seconds. Bearing in mind that the line segment OX represents the path of X 's clock relative to O , it follows that the length of OX in this graph is the time, t_R , as recorded by O , for X 's clock to reach its position relative to O after time t .



It follows by Pythagoras's Theorem that

$$t_R^2 = t^2 + (s/c)^2 \quad (1)$$

Now the speed, v , of X relative to O (that is, relative to O 's clock) is given by $v = s/t_R$, so that Equation (1) becomes

$$t_R^2 = t^2 + (vt_R/c)^2$$

Then after re-arranging this becomes,

$$t = (1 - v^2/c^2)^{1/2} t_R \quad (2)$$

This equation implies that O 's time is dilated relative to X 's time and that a moving clock runs *slower* relative to a stationary observer's clock. Equation (2) is the standard formula for *time-dilation* which appears in Special Relativity. Our derivation shows that, observationally, there are two distinct measures of the speed of relative motion. One is the rate of change of observational distance relative to the time, t_R , as registered by the observer of the motion, known as the **observational time**, and the other is the rate of change of that same distance relative to the time, t , that the observer sees in his telescope/spectroscope registered by the moving body itself, known as the **proper time**. Remember that t is *independent* of any observer. In contrast, the observational time, t_R , is clearly observer dependent.

2. The cone model

Let t , s/c and t_R all be treated as variables in Equation (1). Then Equation (1) is the equation of a *right-circular cone* in three dimensions, relative to three mutually perpendicular time axes, t , s/c and t_R . As such, this conic surface represents a three-dimensional graph of time-dilation, as shown on the front cover of 'Light Speed ...' For any fixed value of t_R , Equation (1) is the equation of a *circle* of radius t_R , centred at the origin. Such circles are obtained from the *end* elevation of the cone. On the other hand, if t is fixed and the time-measure s/c is plotted against the time-measure t_R , the result is a *rectangular-hyperbola* with equation

$$t_R^2 - (s/c)^2 = t^2$$

Such hyperbolæ are obtained from the *side* elevation of the cone. Any such curve is asymptotic to the lines with equations $s = \pm ct_R$.

[*End of recap*]

3. The non-paradox of time-delayed instantaneity

Mach's contribution to Physics provided not only the foundations for Special Relativity but also for the so-called 'Copenhagen' interpretation of Quantum Physics, established by Neils Bohr in the 1920s. This interpretation does not allow any separation between observer and the system observed, regardless of distance, a condition known as *non-locality*. This implies that some type of *instantaneous action-at-a-distance* takes place. In May 1935, a paper written by Einstein in collaboration with Boris Podolski (1896-1966) and Nathan Rosen (1909-1995), appeared. This paper challenged Bohr and his followers on the grounds of Einstein's assumption of *locality* or separateness, according to which it is nonsense to expect that determining the state of a particle at one place is tantamount to determining, *immediately*, the state of another particle somewhere else, regardless of distance. Such instantaneous action-at-a-distance was forbidden by Special Relativity, according to which 'nothing can travel faster than the speed of light'. Einstein and his co-workers argued that, at the very least, Bohr's assertion of non-locality must be incomplete in some sense. (Einstein disparagingly called this alleged quantum instantaneous connection 'spooky action-at-a-distance'.)

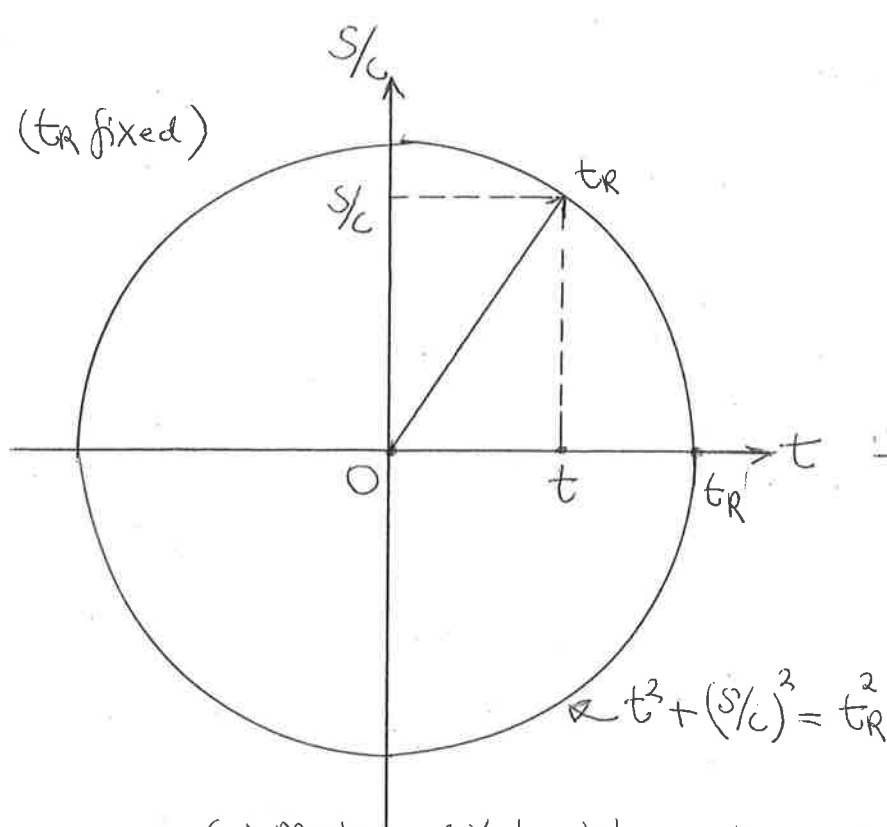
[*I will discuss the Copenhagen approach to Quantum Physics and the EPR paper in greater detail in a later lecture.*]

This clash between Bohr and Einstein continued unresolved for almost thirty years and became known as the 'Einstein-Podolski-Rosen (EPR) controversy'. Then in 1964, John Bell (1928-1990) of CERN (The European Organisation for Nuclear Research, Geneva) suggested a way of deciding, on experimental grounds, whether or not Bohr's instantaneous action-at-a-distance existed. Such experiments were performed in 1978 at Berkeley, USA, by John Clauser and his team, and in 1982, by Alain Aspect and his group in Paris. The results of these experiments supported

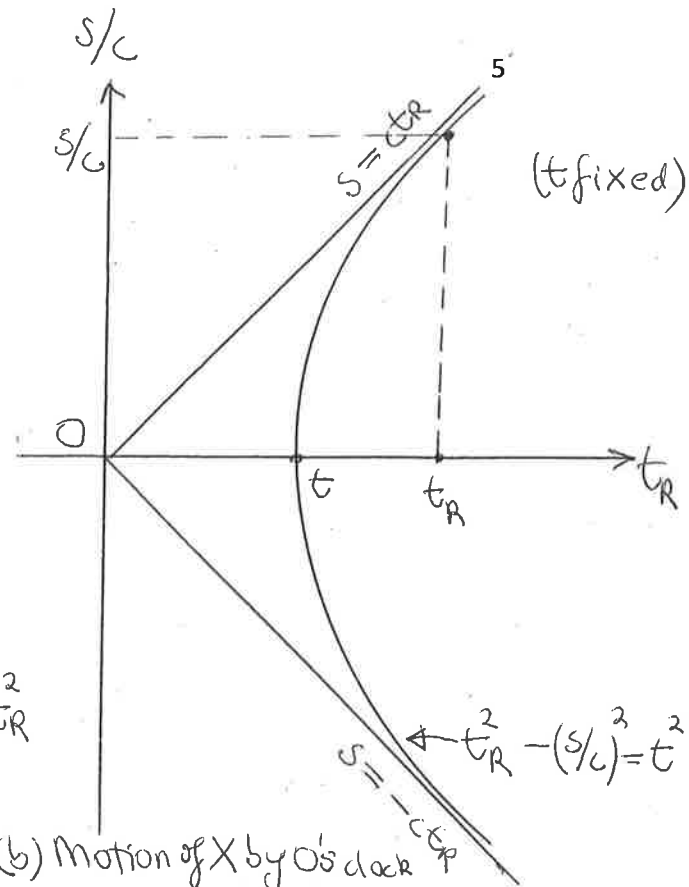
Bohr's contention against Einstein's. However, the controversy over what is now known as the 'EPR paradox' continues to rage. The 'paradox' is that both Bohr's and Einstein's positions appear to be confirmed experimentally. There is, therefore, an apparent contradiction within the context of any of the standard models, whereby on the one hand, Relativistic Physics implies that 'nothing can travel faster than light', while on the other, Quantum Physics implies that distant interactions can be instantaneous.

Instead of seeing this paradox as the natural *reduction ad absurdum* of the 'speed' interpretation of c , some theories have taken the experimental evidence of instantaneous action-at-a-distance as having established the existence of some acausal, 'telepathic' influence passing between particles in advance of the 'speed of light'. The fact that these theories do not define the sort of instantaneity they ascribe to these actions compounds the problem.

The relevance of the cone model to this action-at-a-distance debate is as follows. There are two distinct speeds associated with the uniform motion of body X relative to an observer O . The first is what we can describe as the 'Newtonian' speed and is s/t , where s is the distance travelled by X in proper time t . This speed is unbounded. The second is what can be described as the 'Einsteinian' speed and is s/t_R , where t_R is the time taken for X to travel the distance s relative to O 's clock. Here, $s/t_R < c$. Our conic representation of time-dilation clearly shows how the Newtonian speed-unlimited graph of physical motion (the *end* elevation of the cone shown in Figure (a) below) and the Einsteinian speed-limited aspect (the *side* elevation of the cone shown in Figure (b) below) *can be equally represented with no conflict whatsoever on the same conic surface*.



(a) Motion of X by X's clock



(b) Motion of X by O's clock

Our model also shows that what is represented in the end elevation as a theoretically unbounded Newtonian speed appears in the side elevation as the Einsteinian 'speed limit' c , this speed limit appearing as an asymptote to the graphs representing material particle motion. This clearly demonstrates that although the constant c is the least upper bound of all relative speeds, it is not a speed as such.

To the question, then, of whether action-at-a-distance is instantaneous or time-delayed, the answer, as far as light is concerned, is plainly that it is *both*. This can also be seen from Equations (1) and (2). If $v = c$ in Equation (2), then $t = 0$, so that light-action is instantaneous in this sense, whereas $t = 0$ in Equation (1) gives $t_R = s/c$ as the delayed time ascertained by observation and experiment. There is no contradiction here, since the instantaneity takes place in one observational dimension, whereas the time-delay takes place in a different observational dimension.

The time-dilation formula, Equation (2), tells us then, in effect, that in the case of light interaction, every quantum of the energy that 'starts out' from a source A 'arrives' instantly at a recipient B ; that is, in its own, intrinsic, proper time $t = 0$. In that case, the relative or observationally delayed time $t_R = s/c$, whence $s/t_R = c$ for any disembodied energy such as light. However, it needs to be stressed that despite this making the signal *appear* to have this 'speed' c , the time-delay t_R over the distance s is not a speed but a pure ratio of conventional units of metres to seconds in the measuring

of distance. In this way, the instantaneous light-action can be considered to be a single quantum event common to both source and observer. This was described by Gilbert Lewis (1875-1946) in 1926 as proper time instantaneous ‘quantum touching’. Indeed, logic tells us that the transportation of energy between atoms has to be instantaneous, regardless of distance, since otherwise conservation of energy is violated.

At first sight, this double aspect of light-action appears puzzling. What we are really saying here is that an action may have both time-delayed and instantaneous *components*, without there being any logical contradiction. The action in a movie testifies to this. In a movie, the instantaneous connection between the objects in the still frames is analogous to the ‘instantaneous’ component, t , of atomic interaction. The analogy of the observationally time-delayed component, t_R , is the sequencing of those still frames in the running of the film.

To summarise, we claim that Normal Realism is unique among the current approaches to modern Physics in that by removing the last relic of Cartesian dualism from Relativity (the speed interpretation of c), it eradicates the notorious conflict between Relativity and Quantum Theory over the issue of direct (*i.e.*, instantaneous) *versus* delayed atomic interaction.

4. The Doppler effect

The (optical) Doppler effect is a well-known observed fact. This concerns the apparent change in frequency of a moving source, as first studied by Christian Doppler (1803-1853) in 1842. The *relativistic* effect is that when a light source is receding from an observer, its light-spectrum is shifted towards the *red*, whereas when it is approaching the observer, its light-spectrum is shifted towards the *blue*. I will now demonstrate how this phenomenon is a direct consequence of the time-dilation formula, Equation (2).

Let X be a light source, travelling at speed v in a fixed direction, relative to an observer O . Initially, we shall assume that X is moving *away* from O . Let the very small period of time between successive ‘pulses’ from the light

source be δt , as recorded by a clock moving with the source X . This time will be recorded by O 's clock as δt_R , where from Equation (2),

$$\delta t_R = (1 - v^2/c^2)^{-1/2} \delta t. \quad (3)$$

However, in that time, the light source X has increased its distance from O by $v\delta t_R$ [*distance = speed $\times$ time*]. Hence, as recorded by O , the next 'pulse' arrives at O at time

$$\delta t_R^* = \delta t_R + \frac{v\delta t_R}{c} = (1 + v/c)\delta t_R$$

It then follows from Equation (3) that

$$\delta t_R^* = (1 + v/c)(1 - v^2/c^2)^{-1/2} \delta t \quad (4)$$

The times δt and δt_R^* are, respectively, inversely proportional to the *proper frequency*, ν , of the source as recorded at the source X , and the *observed frequency*, ν_R , of the source as recorded by O . It then follows by equation (4) that

$$\begin{aligned} \frac{\nu_R}{\nu} &= \frac{\delta t}{\delta t_R^*} = \frac{\delta t}{(1 + v/c)(1 - v^2/c^2)^{-1/2} \delta t} \\ &= \frac{1}{(1 + v/c)(1 + v/c)^{-1/2}(1 - v/c)^{-1/2}} \end{aligned}$$

and hence,

$$\nu_R = \frac{(1 - v/c)^{1/2}}{(1 + v/c)^{1/2}} \nu \quad (5)$$

Since the light source X is receding from the observer O , the speed v in equation (5) is positive, and so it follows that $\nu_R < \nu$. [$1 - v/c < 1 + v/c$] It then follows that the light is red-shifted by the relative motion of the source. If, on the other hand, X is approaching O , so that v is taken to be negative in equation (5), it follows that $\nu_R > \nu$. [$1 + v/c > 1 - v/c$ for positive v] In this case, the light is blue-shifted by the relative motion of the source.

The simple scenario presented here can be generalised to more complicated situations, but so far this argument should be sufficient to indicate the versatility of the time-dilation formula.

5. The 'twins paradox'

A famous controversy which has surrounded the time-dilation formula, Equation (2), from soon after its first appearance, is the so-called 'twins paradox' or 'clock paradox'. Consider twins each of whom carries a clock (any type of time measuring device). Initially, they are together on Earth and synchronize their clocks. One twin, O , say, remains on Earth, while the other, X , say, sets off in a space-craft, which after leaving Earth, travels at very high constant speed, in a fixed direction, to visit a distant planet. The space-traveller, X , then returns to Earth at the same high constant speed, in a fixed direction, relative to O and meets up again with O . Ignoring the accelerations and decelerations on take-off and touch-down, it follows by Equation (2) that X 's clock will have recorded a shorter passage of time than O 's clock relative to O , both on the outward and return journeys. It follows that if the twins are the same age before the space-traveller's journey, then when they meet up again, the space-traveller will be younger than the other twin, since each observer ages at the same rate relative to his own clock.

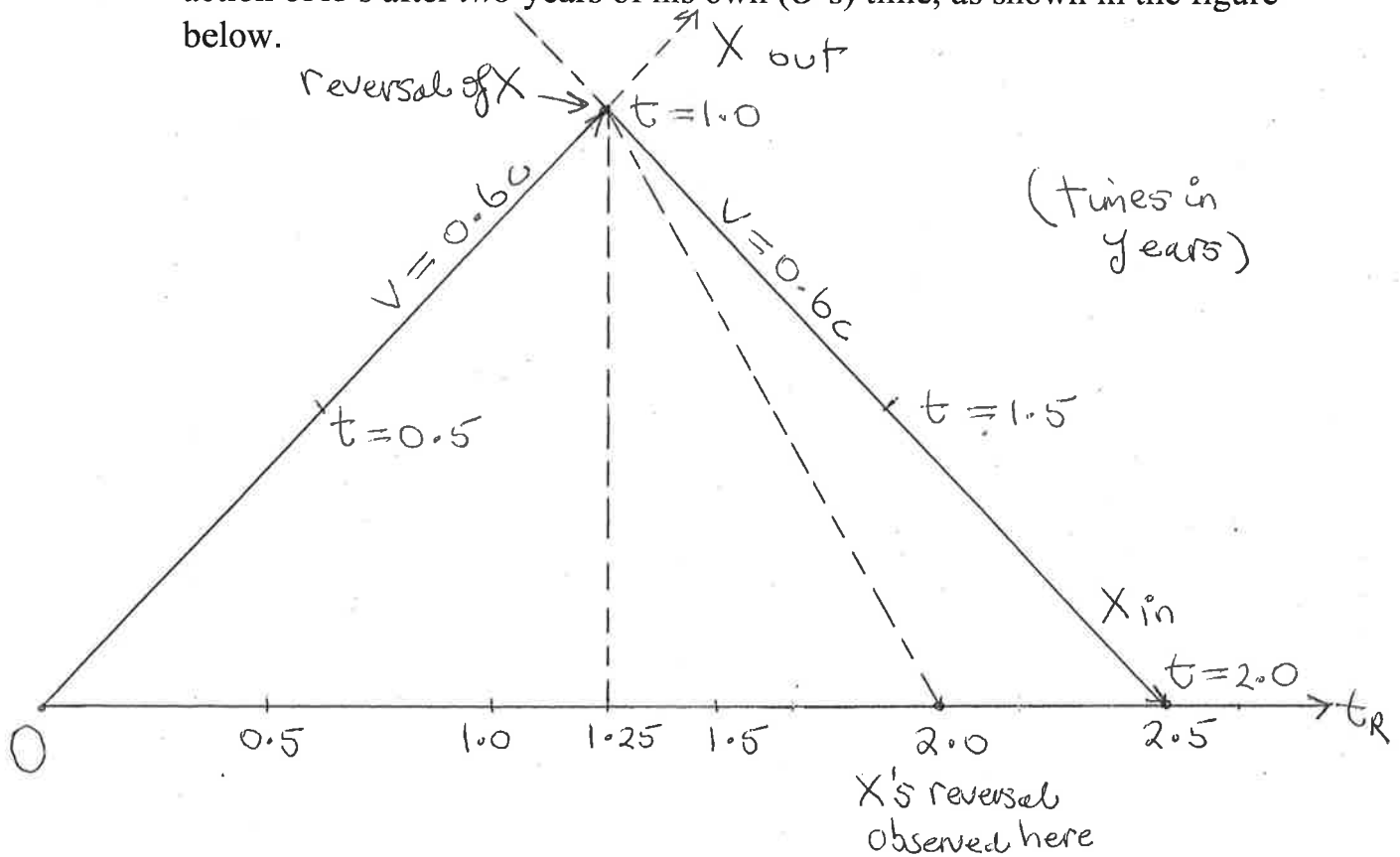
The alleged 'paradox' is this. On the face of it, it appears that X could claim with equal right that it was he who remained where he was, while it was O who went on the round trip, with the consequence that O should be the younger when they meet again. However, there is no such paradox, because it must be remembered that Equation (2) applies only to relative *uniform* motion. In order to make his trip, the space-traveller X must undergo an initial period of acceleration and when he approaches the distant planet, he must decelerate to land. The same is true on the return journey to Earth. These periods of acceleration and deceleration, no matter how short compared to the length of the whole journey, have definitely observable and measurable physical (for example, visceral) effects. Hence, X is not in an inertial frame for the whole journey, in contrast to O , and so O 's and X 's observational frames are not physically or mathematically interchangeable.

This is essentially the argument provided by Einstein in his correspondence with Viv in 1954. Although Equation (2) applies for most of X 's journey relative to O , the periods of acceleration and deceleration of X , no matter how short compared to the length of the whole journey, destroy the otherwise symmetrical situation between O and X . For these short periods, X will be very aware of the physical effects on his body produced by the non-uniform motion (in the form of 'g forces'), whereas O will detect no such effects. A good analogy for the fact that very short periods of non-uniform motion have a huge impact on the whole journey has been provided by Bondi. Imagine travelling in a car at constant speed along a straight road. Suppose at some point, the car turns right and continues to travel at constant speed along another straight road. The negligible period of non-uniform motion, when the car turns right, has a huge effect on the car's destination!

Let us examine this asymmetry between the twins O and X in more detail. In order to remove as many effects due to non-uniform motion as possible, imagine O and X being two astronauts somewhere in deep space, far away from any possible gravitational effects. As before, X moves away from O at constant speed, v , in a fixed direction, then reverses direction (and so undergoes non-uniform motion) and returns to O at the same constant speed, along the same direction. Remember that any precept that there is some underlying 'God's-eye-view' that can be taken of the recordings of O and X together must be abandoned. In that case, the only way in which the twins can compare their time-readings is for each of them, by means of a telescope, say, to read the other's clock and comparing those readings with the readings of their own clock. To be specific, let us suppose that throughout the out-and-back motion, O and X keep observing each other through telescopes fitted with spectroscopes which record the number of atomic periods counted by each other.

As O and X move apart in space, the ticks of the atomic clock that each twin observes of the other in his telescope are slowed relative to his own due to the extra time s/c incurred by the increasing distance between them. This is the observed Doppler effect, as explained in the last section. (Recall that when a body is receding from an observer, its light-spectrum is shifted towards the red, so that the 'ticks' of the atomic clock appear slowed, whereas when the body is approaching the observer, its light-spectrum is shifted towards the blue, so that its 'atomic ticks' appear quickened.) Suppose that O and X move steadily apart at such a speed that the ticks of

each one's clock, as viewed by the other, are twice as long as his own. Suppose also that X reverses direction after one year, as recorded by his (X 's) atomic clock. We know that due to the Doppler lengthening of the ticks of X 's atomic clock in O 's view of X , O will record that reversing action of X 's after *two* years of his own (O 's) time, as shown in the figure below.



In this case, it follows from equation (5), with $v/v_R = 2$, that

$$(1 + v/c) = 4(1 - v/c) \Rightarrow v/c = 3/5$$

Hence, the speed of X relative to O is $v = 0.6c$. Neglecting the time taken for his turn-around, X meets up with O after another year, as recorded by X 's own atomic clock. Meanwhile, during the return journey of X , after the two years already elapsed according to O 's clock, it follows by Equation (5) that O will observe the ticks of X 's atomic clock from then on at the faster (blued) rate of half their standard length, which means that in the one year that X 's atomic clock will have registered in returning to O , O 's atomic clock will have registered a half of that time, that is, six months. It follows, then, that while the total time registered by X 's clock between their leaving and meeting up again is just *two years*, for O it is *two and a half years*. And what is the case for atomic clocks must obviously be the same for all other processes, such as heartbeats, aging-rates and so on. Hence, when they get

back together, the results of the relative motion is that twin O will have aged six months more than twin X .

The situation for O and X is therefore plainly *asymmetrical*, because O counts slower ticks from X for a longer period than X does from O and counts faster ticks from then on for a correspondingly shorter period. This asymmetry is graphically portrayed in the figure. Notice that the time-dilation factor from O 's time to X 's time for the whole journey is 1.25. In this case, Equation (2) verifies a speed of $v = 0.6c$ for X relative to O , on both the outward and return journeys:

$$(1 - v^2/c^2)^{1/2} = 4/5 \Rightarrow v^2/c^2 = 9/25 \Rightarrow v = 3c/5$$

In summary, this analysis reveals that whichever of the twins undertakes a reversing action to bring the two back together switches immediately at that point from counting red-shifted to blue-shifted periods in his spectroscopic readings of the other's clock. Meanwhile, for the other, that reversing action will not be observed immediately. This asymmetry in the period-counting, between the observers, disposes of the so-called paradox.

In the next lecture, I will present further consequences of the time dilation formula, including Lorentz transformations, Minkowski space-time and length contraction, which are standard consequences of Special Relativity.

Anthony D. Osborne, April 2023.